

YUCHEN YANG

Assistant Professor
College of Information Sciences and Technology
The Pennsylvania State University
University Park, PA 16802

✉ yuchen.yang@psu.edu
☎ (814) 863-8320
🌐 <https://yuchenyang.org>
📍 413A Eric J. Barron Innovation Hub

PROFESSIONAL EXPERIENCES

Assistant Professor	2025.07 - Current
The Pennsylvania State University , PA, United States	
Research Assistant for Dr. Yinzhi Cao	2020.03 - 2025.05
The Johns Hopkins University , MD, United States	
Student Associate for Dr. Shao-yuan Lo and Dr. Behzad Dariush	2023.10 - 2024.02
Honda Research Institute , CA, United States	
Research Assistant for Dr. Yuqing Sun	2018.12 - 2019.06
Shandong Univeristy , Shandong, China	
Research Assistant for Dr. Yingjie Tian	2018.06 - 2018.09
Chinese Academy of Sciences , Beijing, China	

EDUCATION

Ph.D. The Johns Hopkins University	2021 – 2025
<i>Computer Science</i>	MD, United States
<i>Advisor: Dr. Yinzhi Cao</i>	
M.S. The Johns Hopkins University	2019 – 2021
<i>Security Informatics</i>	MD, United States
<i>Advisor: Dr. Yinzhi Cao</i>	
B.E. Shandong University	2015 – 2019
<i>Software Engineering</i>	Shandong, China

SELECTED PUBLICATIONS

PEER-REVIEWED CONFERENCE PAPERS (* INDICATES CO-FIRST AUTHORS)

- 2025 **CertPHash: Towards Certified Perceptual Hashing via Robust Training**
Yuchen Yang, Qichang Liu, Christopher Brix, Huan Zhang, Yinzhi Cao
In the proceedings of USENIX Security Symposium (Usenix), 2025
- 2024 **Follow the Rules: Reasoning for Video Anomaly Detection with Large Language Models**
Yuchen Yang, Kwonjoon Lee, Behzad Dariush, Yinzhi Cao, Shao-Yuan Lo
In the Proceedings of the European Conference on Computer Vision (ECCV), 2024
SafeGen: Mitigating Sexually Explicit Content Generation in Text-to-Image Models
Xinfeng Li*, **Yuchen Yang***, Jiangyi Deng*, Chen Yan, Yanjiao Chen, Xiaoyu Ji, Wenyan Xu
In the Proceedings of the ACM Conference on Computer and Communications Security (CCS), 2024
RippleCOT: Amplifying Ripple Effect of Knowledge Editing in Language Models via Chain-of-Thought In-Context Learning
Zihao Zhao, **Yuchen Yang**, Yijiang Li, Yinzhi Cao
In the Findings of the Empirical Methods in Natural Language Processing (EMNLP), 2024
SneakyPrompt: Jailbreaking Text-to-image Generative Models
Yuchen Yang, Bo Hui, Haolin Yuan, Neil Gong, Yinzhi Cao
In the Proceedings of the IEEE Symposium on Security and Privacy (S&P), 2024
Media Coverage (Selected): [MIT Technology Review](#) [🔗](#)
- 2023 **PrivateFL: Accurate, Differentially Private Federated Learning via Personalized Data Transformation**
Yuchen Yang*, Bo Hui*, Haolin Yuan*, Neil Gong, Yinzhi Cao

In the proceedings of USENIX Security Symposium (Usenix), 2023

Fortifying Federated Learning against Membership Inference Attacks via Client-level Input Perturbation

Yuchen Yang, Haolin Yuan, Bo Hui, Neil Gong, Neil Fendley, Philippe Burlina, Yinzhi Cao.

In the proceedings of IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2023

2022 **Addressing Heterogeneity in Federated Learning via Distributional Transformation**

Haolin Yuan*, Bo Hui*, **Yuchen Yang***, Philippe Burlina, Neil Gong, Yinzhi Cao

In the proceedings of the European Conference on Computer Vision (ECCV), 2022

2021 **Practical Blind Membership Inference Attack via Differential Comparisons**

Bo Hui*, **Yuchen Yang***, Haolin Yuan*, Philippe Burlina, Neil Gong, Yinzhi Cao

In the proceedings of Network & Distributed System Security Symposium (NDSS), 2021

PATENT

2024 **System and Method Using Reasoning for Video Anomaly Detection with Large Language Models**

Yuchen Yang, Kwonjoon Lee, Shao-Yuan Lo, Behzad Dariush

Under US Patent Application (Filed), 2024

PROFESSIONAL SERVICES

CONFERENCE/JOURNAL REVIEWING

PC , USENIX Security Symposium	2026
IEEE Symposium on Security and Privacy (S&P)	2026
ACM Conference on Computer and Communications Security (CCS)	2025
ACM Workshop on Adaptive and Autonomous Cyber Defense (AACD)	2024
AEC , IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)	2024
Reviewer , International Conference on Learning Representations (ICLR)	2025
International Conference on Computer Vision (ICCV)	2025
IEEE Transactions on Dependable and Secure Computing (TDSC)	2023, 2024
IEEE Transactions on Information Forensics & Security (T-IFS)	2024
IEEE Computer Security Foundations Symposium (CSF)	2022, 2024
IEEE International Conference on Distributed Computing Systems (ICDCS)	2022

ORGANIZING AND CHAIRING

Session Chair , IEEE Workshop on Deep Learning Security and Privacy (DLSP)	2024
---	------

SELECTED MEDIA COVERAGE

Text-to-image models can be tricked into generating disturbing images, MIT Technology Review	2023.11
AI art generators can be fooled into making NSFW images, IEEE Spectrum	2023.11
Nonsense prompts trick AIs into producing NSFW images, Technology Networks	2023.11
Researchers reveal vulnerabilities in AI models, prompting concerns, Cryptopolitan	2023.11
AI generators can be tricked into making NSFW content, JHU Engineering School News	2023.11
SneakyPrompt: Revealing the vulnerabilities of text-to-image AI, The Johns Hopkins News-Letter	2023.12

INVITED TALKS

"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", Penn State	2025.3
"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", CWRU	2025.2
"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", Rice University	2025.2
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Virginia Tech	2025.2
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", RIT	2025.1
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Monash University	2024.10
"Zero-shot Video Anomaly Detection: Steering LLMs for Rules-based Reasoning", Voxel51	2024.11